



An Improved Plate Deep Energy Method for the Bending, Buckling and Free Vibration Problems of Irregular Kirchhoff Plates

Course: Theory of Plates and Shells

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Prior Art & This Work's Contribution

1 Introduction

Existing neural PDE solvers

- **PINNs** (Raissi et al. 2019); strong form
- **Deep Ritz Method** (Weinan & Yu 2018); variational form
- **Deep Energy Method (DEM)** (Samaniego et al. 2020); energy minimization
- **PhyGeoNet**; finite-difference on irregular domains

Common limitation: penalty method for BCs

This paper introduces

- **Penalty-free** deep energy method
- **R-function** distance functions for Dirichlet BCs on irregular boundaries
- **Virtual nodes** for derivative BCs (clamped edges)
- **Convolution-based finite differences**; faster training, no AD for forward derivatives
- Validated on bending, free vibration & buckling of **irregular Kirchhoff plates**



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Kirchhoff Plate Equations

2 Mechanical Model of Kirchhoff Plate

Strain–Displacement

$$\varepsilon_x = -z \frac{\partial^2 w}{\partial x^2} \quad (1a)$$

$$\varepsilon_y = -z \frac{\partial^2 w}{\partial y^2} \quad (1b)$$

$$\gamma_{xy} = -z \frac{\partial^2 w}{\partial x \partial y} \quad (1c)$$

Constitutive

$$\sigma_x = \frac{E}{1-\nu^2} (\varepsilon_x + \nu \varepsilon_y) \quad (2a)$$

$$\sigma_y = \frac{E}{1-\nu^2} (\varepsilon_y + \nu \varepsilon_x) \quad (2b)$$

$$\tau_{xy} = \frac{E}{2(1+\nu)} \gamma_{xy} \quad (2c)$$

Bending & Twisting Moments

$$M_x = -D \left(\frac{\partial^2 w}{\partial x^2} + \nu \frac{\partial^2 w}{\partial y^2} \right) \quad (3)$$

$$M_y = -D \left(\frac{\partial^2 w}{\partial y^2} + \nu \frac{\partial^2 w}{\partial x^2} \right) \quad (4)$$

$$M_{xy} = -(1-\nu)D \frac{\partial^2 w}{\partial x \partial y} \quad (5)$$

$$\text{where } D = \frac{Et^3}{12(1-\nu^2)} \text{ (bending stiffness)}$$



Kirchhoff Plate Equations

2 Mechanical Model of Kirchhoff Plate

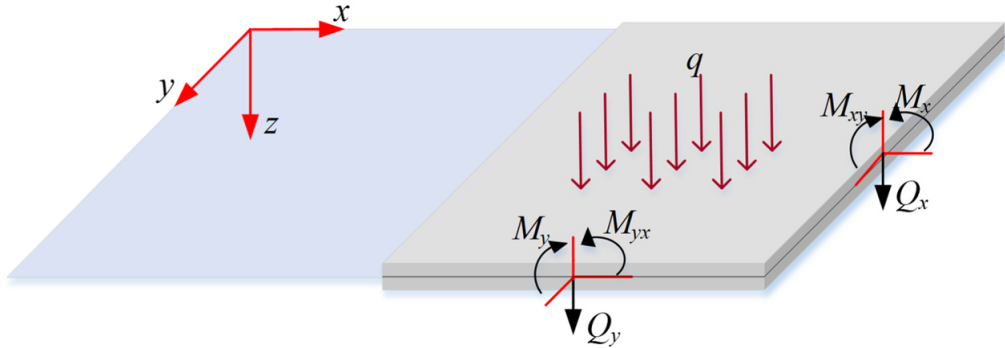


Figure 1: Kirchhoff thin plate model (Cartesian coordinate system).



Plate Bending

2 Mechanical Model of Kirchhoff Plate

$$\Pi = U + W_p.$$

The strain energy

$$U = -\frac{1}{2} \iint_A \left(M_x \frac{\partial^2 w}{\partial x^2} + M_y \frac{\partial^2 w}{\partial y^2} + 2M_{xy} \frac{\partial^2 w}{\partial x \partial y} \right) dA.$$

The potential energy

$$W_p = - \left[\iint_A q(x, y) w dA + \sum_i P_i w_i + \sum_j M_j \theta_j \right],$$



Free Vibration of Plates

2 Mechanical Model of Kirchhoff Plate

Rayleigh's principle

$$K_{max} = U_{max}.$$

maximum kinetic energy (harmonic vibrations)

$$K_{max} = \frac{\omega^2}{2} \iint_A \rho w^2(x, y) \, dA$$

and the maximum strain energy

$$U_{max} = \frac{1}{2} \iint_A D \left\{ \left(\nabla^2 w \right)^2 + 2(1 - \nu) \left[\left(\frac{\partial w}{\partial x \partial y} \right)^2 - \frac{\partial^2 w}{\partial x^2} \frac{\partial^2 w}{\partial y^2} \right] \right\} \, dA.$$



Plate Buckling

2 Mechanical Model of Kirchhoff Plate

The critical load parameter (general energy criterion)

$$\lambda_{\text{cr}} = \frac{D \iint_A \left\{ \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} \right)^2 - 2(1 - \nu) \left[\frac{\partial^2 w}{\partial x^2} \frac{\partial^2 w}{\partial y^2} - \left(\frac{\partial^2 w}{\partial x \partial y} \right)^2 \right] \right\} dx dy}{\iint_A \left[N'_x \left(\frac{\partial w}{\partial x} \right)^2 + N'_y \left(\frac{\partial w}{\partial y} \right)^2 + 2N'_{xy} \frac{\partial w}{\partial x} \frac{\partial w}{\partial y} \right] dx dy}$$

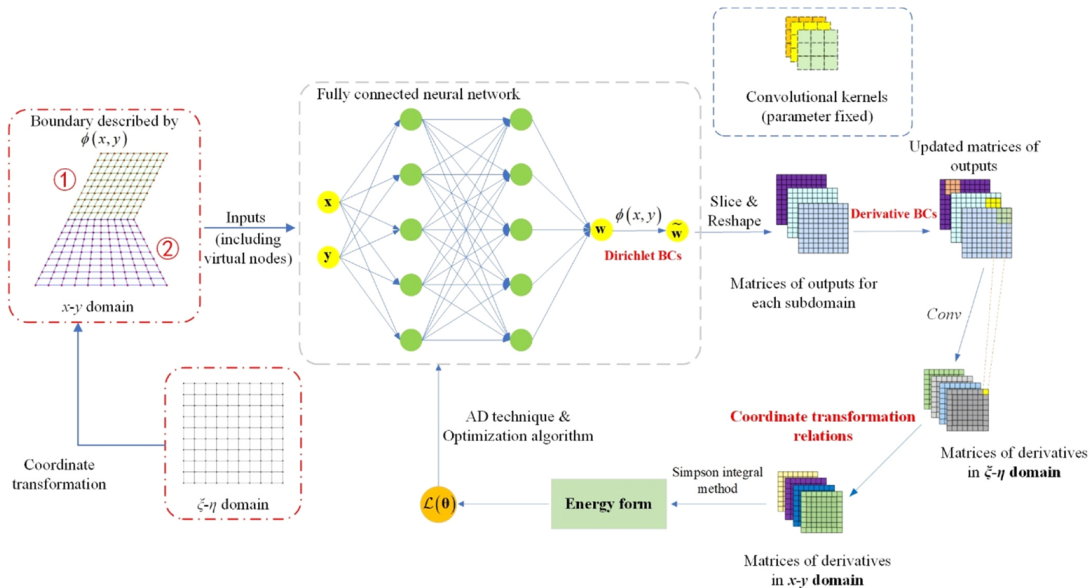




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Step 1: Domain Decomposition & Coordinate Mapping

3 Domain Decomposition & Coordinate Transformation

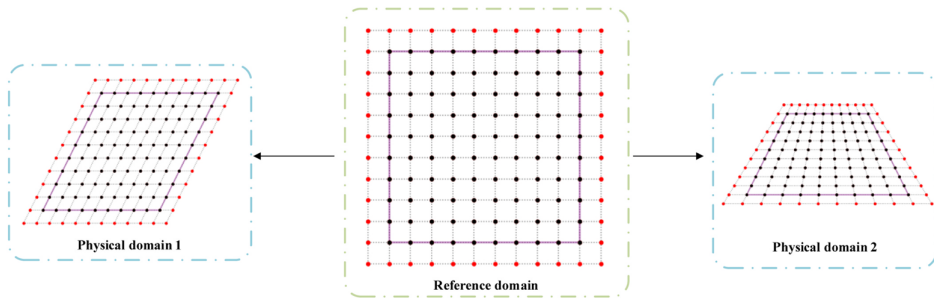


Figure 2: Project the reference domain to physical domains



Step 1: Domain Decomposition & Coordinate Mapping

3 Domain Decomposition & Coordinate Transformation

- Each physical domain mapped to a **unit square reference domain** $[0, 1]^2$
- Coordinate transformation $(\xi, \eta) \rightarrow (x, y)$ obtained by **interpolation**
- Chain rule gives 1st- and 2nd-order derivative relations:

$$x(\xi, \eta) = \sum_{i=1}^4 N_i(\xi, \eta) x_i$$

$$y(\xi, \eta) = \sum_{i=1}^4 N_i(\xi, \eta) y_i$$

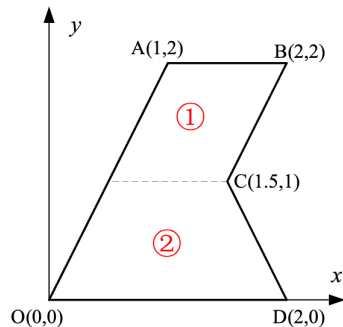


Figure 3: Irregular plate.



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Step 2: Data Generation & Grid Setup

4 Data Generation

- Training nodes generated **uniformly** in the reference domain, then mapped to physical domain
- Input data **fixed** during training
- Network output reshaped to matrix of size (L, M) for convolution
- **Virtual nodes** added as padding: matrix becomes $(L+2, M+2)$
 - Prevents loss of boundary derivatives after convolution
 - Also used to enforce derivative BCs (Step 4)

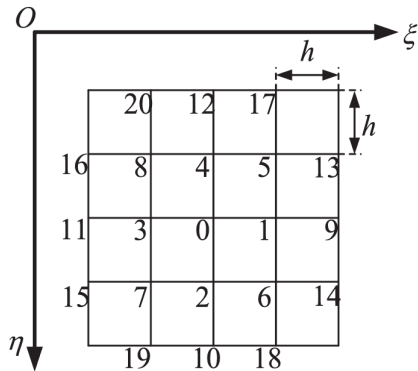


Figure 4: Node number of the difference grid in the reference domain.



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Step 3: Distance Function & BCs

5 Distance Function (R-Function)

Multiply NN output by distance function:

$$W(\mathbf{x}; \boldsymbol{\theta}) = \phi(\mathbf{x}) \mathcal{N}(\mathbf{x}; \boldsymbol{\theta})$$

$\phi(\mathbf{x}) = 0$ on the Dirichlet boundary $\Rightarrow$ BC satisfied **exactly**, **no penalty term**. **R-function theory** constructs ϕ for complex domains:

$$x_1 \wedge_{\alpha} x_2 = \frac{1}{1 + \alpha} \left(x_1 + x_2 - \sqrt{x_1^2 + x_2^2 - 2\alpha x_1 x_2} \right)$$

$$x_1 \vee_{\alpha} x_2 = \frac{1}{1 + \alpha} \left(x_1 + x_2 + \sqrt{x_1^2 + x_2^2 - 2\alpha x_1 x_2} \right)$$



Step 3: Distance Function & BCs

5 Distance Function (R-Function)

Annular Plate; Direct

$$\omega_1 = a^2 - x^2 - y^2$$

$$\phi_1 = \omega_1$$

$$\boxed{\phi = (a^2 - x^2 - y^2)^2}$$

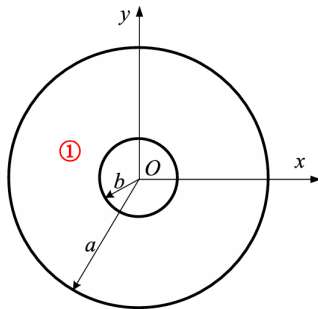


Figure 5: Annular Plate



Step 3: Distance Function & BCs

5 Distance Function (R-Function)

L-Shaped Plate; R-Function

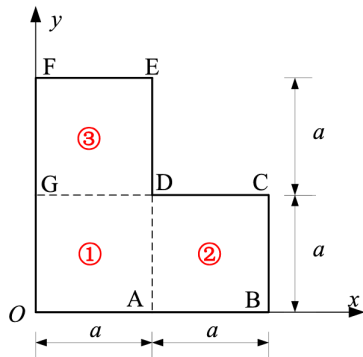


Figure 6: L-Shaped Plate

$$\omega_1 = x(x - 2a)$$

$$\omega_2 = y(y - 2a)$$

$$\omega_3 = (a - x)$$

$$\omega_4 = (a - y)$$

$$\phi_1 = (\omega_1 \wedge \omega_2) \wedge (\omega_3 \vee \omega_4)$$



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Step 4: Derivative BCs via Virtual Nodes

6 Derivative BCs & Virtual Nodes

For **clamped edges**, after $w = 0$ is imposed by ϕ , we still need $\partial w / \partial n = 0$.

On edge $\xi = 1$: $w = 0$ implies $\partial w / \partial \eta = 0$, so

$$\frac{\partial w}{\partial \xi} = 0 \quad \Longrightarrow \quad \left(\frac{\partial f}{\partial \xi} \right)_{\xi=1} \approx \frac{1}{2h} (w_{l_1} - w_r) = 0$$

Two schemes for virtual node value w_r :

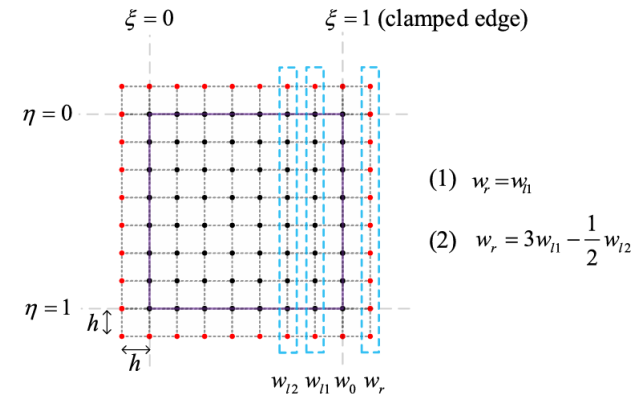
1. *1st-order*: $w_r = w_{l_1}$
2. *Higher-order* (cubic fit):

$$w_r = 3w_{l_1} - \frac{1}{2}w_{l_2}$$



Step 4: Derivative BCs via Virtual Nodes

6 Derivative BCs & Virtual Nodes



$$(1) \quad w_r = w_{l1}$$

$$(2) \quad w_r = 3w_{l1} - \frac{1}{2}w_{l2}$$

Figure 7: Diagram of the enforcement of clamped boundary conditions at edge $\xi = 1$ within the reference domain.

- Collocation nodes
- Virtual nodes
- Boundary line



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Step 5: Convolutional Finite Differences

7 Derivatives in the Physical Domain

Central-difference stencils (mesh interval h):

$$\left(\frac{\partial f}{\partial \xi}\right)_0 = \frac{f_1 - f_3}{2h}, \quad \left(\frac{\partial^2 f}{\partial \xi^2}\right)_0 = \frac{f_1 - 2f_0 + f_3}{h^2}$$

$$\left(\frac{\partial^2 f}{\partial \xi \partial \eta}\right)_0 = \frac{f_6 + f_8 - f_5 - f_7}{4h^2}$$

0	0	0
-1	0	1
0	0	0

$\times \frac{1}{2h}$

0	-1	0
0	0	0
0	1	0

$\times \frac{1}{2h}$

Figure 8: Convolutional kernels for $\left(\frac{\partial f}{\partial \xi}\right)_0$ (left) and $\left(\frac{\partial f}{\partial \eta}\right)_0$ (right) at node 0.



Step 5: Convolutional Finite Differences

7 Derivatives in the Physical Domain

$$\begin{array}{|c|c|c|} \hline 0 & 0 & 0 \\ \hline 1 & -2 & 1 \\ \hline 0 & 0 & 0 \\ \hline \end{array} \times \frac{1}{h^2} \quad
 \begin{array}{|c|c|c|} \hline 0 & 1 & 0 \\ \hline 0 & -2 & 0 \\ \hline 0 & 1 & 0 \\ \hline \end{array} \times \frac{1}{h^2} \quad
 \begin{array}{|c|c|c|} \hline 1 & 0 & -1 \\ \hline 0 & 0 & 0 \\ \hline -1 & 0 & 1 \\ \hline \end{array} \times \frac{1}{4h^2}$$

Figure 9: Convolutional kernels for $\left(\frac{\partial^2 f}{\partial \xi^2}\right)_0$ (left), $\left(\frac{\partial^2 f}{\partial \eta^2}\right)_0$ (middle), and $\left(\frac{\partial^2 f}{\partial \xi \partial \eta}\right)_0$ (right) at node 0.



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Step 6: Loss Functions

8 Loss Functions

Bending

$$\mathcal{L}_{\text{bend}} = \sum_{i=1}^N \Pi_i(\mathbf{x}; \boldsymbol{\theta})$$

$$\Pi_i = \frac{1}{2} \int_{A_i} \mathbf{k}_i^T \mathbf{D} \mathbf{k}_i dA - \int_{A_i} qW dA$$

curvature vector $\mathbf{k} = [-W_{xx}, -W_{yy}, -2W_{xy}]^T$

Free Vibration

$$\mathcal{L}_{\text{vib}} = \omega^2(\mathbf{x}; \boldsymbol{\theta}) = \frac{2 \sum_i U_i^{\max}}{\sum_i \iint_{A_i} \rho t W^2 dA}$$

Minimising ω^2 yields the **fundamental frequency**.

Buckling

$$\mathcal{L}_{\text{buck}} = \lambda(\mathbf{x}; \boldsymbol{\theta})$$

$$\lambda = \frac{\frac{1}{2} \sum_i \iint_{A_i} D [(\nabla^2 W)^2 + \dots] dA}{-\frac{1}{2} \sum_i \iint_{A_i} [N_x W_x^2 + N_y W_y^2 + 2N_{xy} W_x W_y] dA}$$

Integration uses 2-D **Simpson's rule**:

$$\mathbf{W}_{\text{simp}}^p = \boldsymbol{\Phi} \odot \mathbf{W}_{\text{simp}}^r$$

($\boldsymbol{\Phi}$ = Jacobian determinants at nodes)



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Step 7: Network Architecture & Optimization

9 Optimization Algorithm

Convergence criterion:

$$\text{conv} = \frac{\sum_i (W_i^* - W_i)^2}{\sum_i (W_i^*)^2} \leq 10^{-6}$$

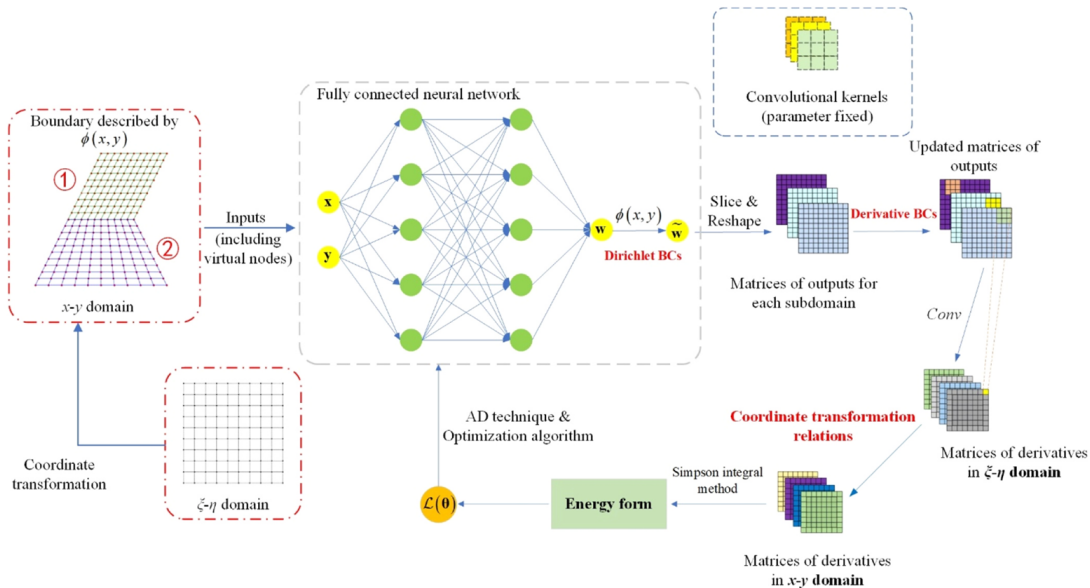




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Example 1: Bending of Irregular Plate

10 Numerical Examples

Setup: Mixed BCs (SS: OA, AB, OD; Clamped: BC, CD), $E = 10.92 \times 10^6$ Pa, $\nu = 0.3$, $t = 0.01$ m, $q_0 = 1$ N/m², 2 subdomains

N_{int}	N_l	N_n	w_c ($\times 10^{-3}$ m)	e_1
$29^2 \times 2$	5	20	-6.155	-0.49%
$35^2 \times 2$	5	20	-6.160	-0.41%
$41^2 \times 2$	5	30	-6.163	-0.35%
$41^2 \times 2$	5	40	-6.189	+0.07%

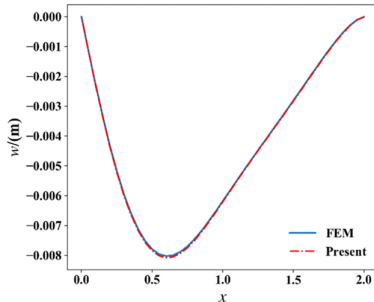
FEM (ABAQUS, 4068 STRI3 elements): $w_c^* = -6.185 \times 10^{-3}$ m

Memory stable (≈ 178 – 183 MiB) regardless of node count.

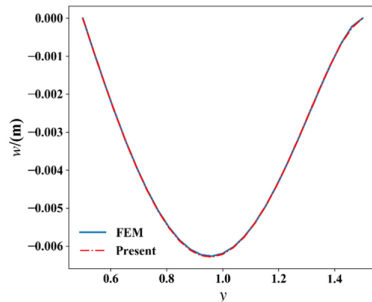


Example 1: Bending of Irregular Plate

10 Numerical Examples



(a) Deflections along axis $x=1m$.



(b) Deflections along axis $y=1m$.

Figure 11: Comparison of deflections given by the presented method($N_l=5$, $N_n=30$, $41 \times 41 \times 2$ training nodes) and the FEM.



Example 2: Bending of Triangular Plate

10 Numerical Examples

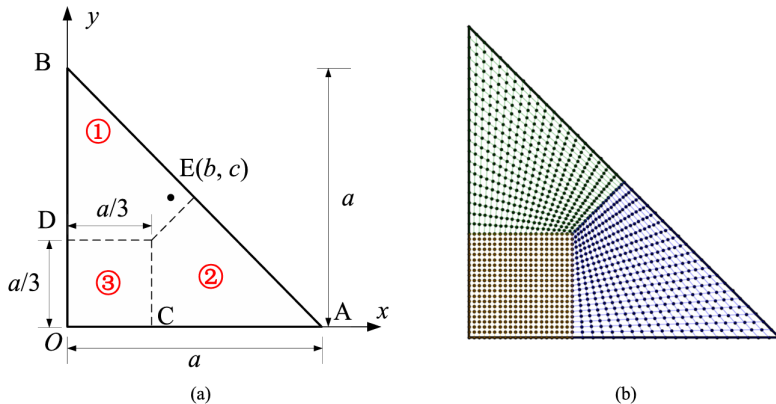


Figure 12: Triangular plate (a) and difference grids (b).



Example 2: Bending of Triangular Plate

10 Numerical Examples

Isosceles right triangle, $a = 1.5$ m, $P_0 = 1$ N at point $E(b, c)$

Distance function: $\phi = xy(y + x - a)$, 3 subdomains, $3 \times 21 \times 21$ nodes

Simply supported vs. theory (dimensionless $\bar{w} = 10^3 Dw/(P_0 a^2)$):

(b, c)	Present	Theory	e_1
$(a/3, a/3)$	4.277	4.268	0.21%
$(a/2, a/3)$	2.554	2.549	0.22%
$(a/6, a/6)$	2.653	2.658	-0.19%

Mixed BC vs. FEM (5379 STRI3 elements):

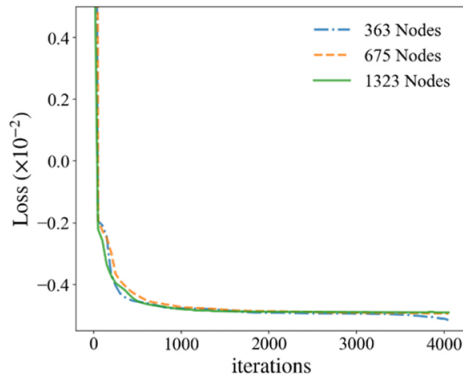
(b, c)	Present	FEM	e_1
$(a/3, a/3)$	3.057	3.046	0.38%
$(a/6, a/6)$	1.882	1.850	1.72%

Present method **converges faster** than DEM; memory usage comparable.

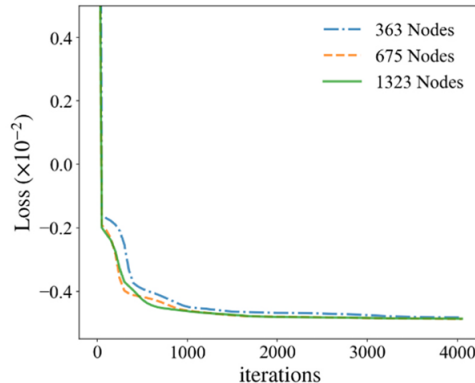


Example 2: Bending of Triangular Plate

10 Numerical Examples



(a) Loss convergence history of the presented method



(a) Loss convergence history of the DEM

Figure 13: Comparison of convergence histories between the presented method and DEM.



Example 3: Bending of L-Shaped Plate

10 Numerical Examples

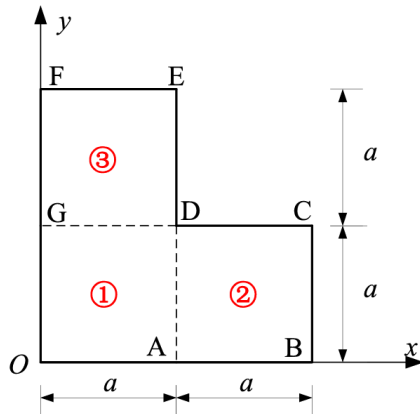


Figure 14: L-shaped plate.



Example 3: Bending of L-Shaped Plate

10 Numerical Examples

Uniform load $q_0 = 1 \text{ N/m}^2$, $a = 0.5 \text{ m}$, 3 subdomains, $3 \times 41 \times 41$ nodes

Two boundary conditions tested:

1. All edges **simply supported**: $\phi_1 = (\omega_1 \wedge \omega_2) \wedge (\omega_3 \vee \omega_4)$
2. All edges **clamped**: $\phi_2 = \phi_1^2$

Results along $\bar{y} = 0.25$ compared with ABAQUS (3850 STRI3 elements)

$\Rightarrow$ **Excellent agreement** for both BCs



Example 3: Bending of L-Shaped Plate

10 Numerical Examples

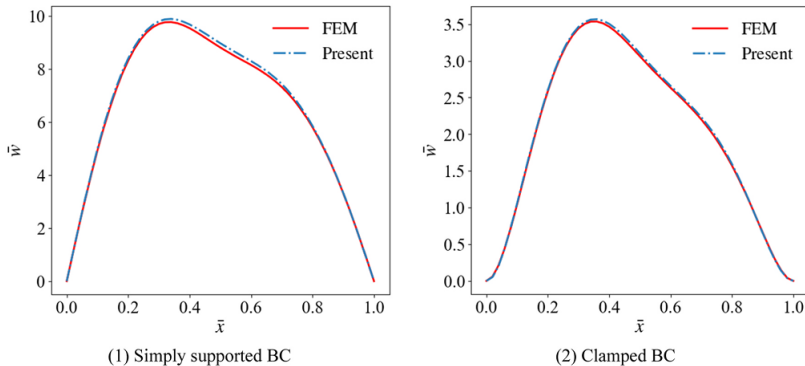
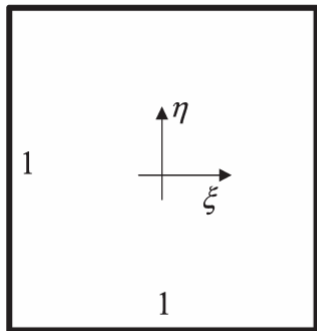


Figure 15: Dimensionless deflections along the axis $y = 0.25$.

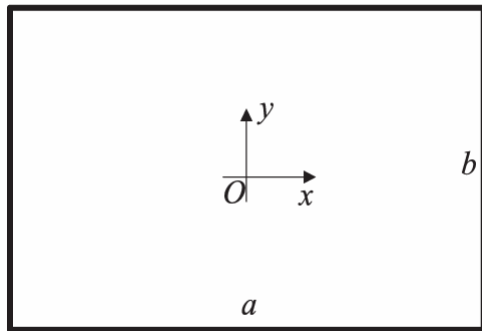


Example 4: Free Vibration of Plates

10 Numerical Examples



Reference domain



Physical domain

Figure 16: Reference domain and physical domain for rectangular plate.



Example 4: Free Vibration of Plates

10 Numerical Examples

Rectangular plates (SSSS): dimensionless frequency $\lambda' = \omega b^2 / (2\pi) \sqrt{\rho t / D}$:

Method	$a/b = 1$	1.5	2	3
Present	3.138	2.267	1.962	1.744
Theory	3.142	2.269	1.964	1.745

Annular plates (free inner, clamped outer) fundamental frequency (Hz):

b/a	Present	FEM	e
0.1	1.631	1.656	-1.50%
0.3	1.810	1.815	-0.24%
0.6	4.086	4.054	+0.79%

Distance function: $\phi = (a^2 - x^2 - y^2)^2$



Example 4: Free Vibration of Plates

10 Numerical Examples

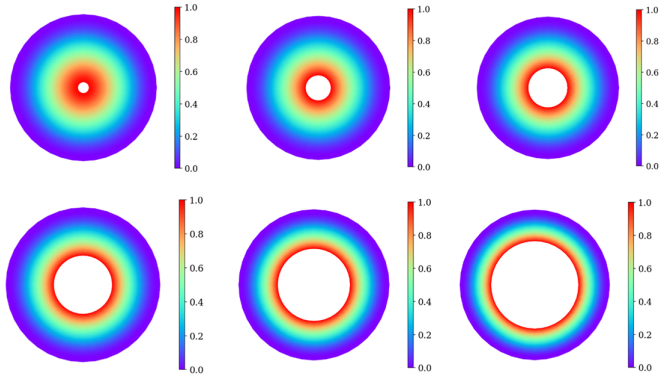


Figure 17: Fundamental mode shapes given by presented method.



Example 5: Buckling of Skew & Irregular Plates

10 Numerical Examples

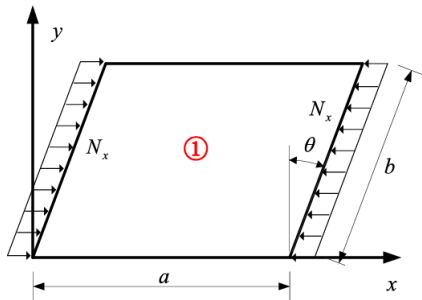


Figure 18: Skew plate.



Example 5: Buckling of Skew & Irregular Plates

10 Numerical Examples

Skew plates buckling factor $k = b^2\lambda/\pi^2$, simply supported:

Method	0°	15°	30°	45°
Present (41 × 41)	3.998	4.399	6.062	10.89
Wang (1992)	4.00	4.44	6.19	10.60
Durvasula	4.00	4.48	6.41	12.30

Irregular plates (symmetric, clamped edges 1&2, free otherwise):

α	Present (Appr. 1)	ABAQUS (STR13)
45°	1.836	1.825
60°	1.002	0.998
80°	0.068	0.670

Good agreement; instability at 80° with Approach 1 due to high skewness — subdomain decomposition recommended.



Example 5: Buckling of Skew & Irregular Plates

10 Numerical Examples

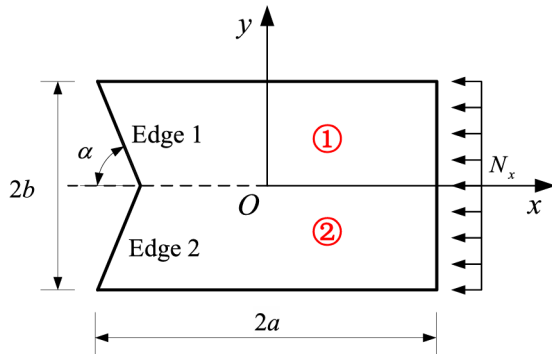


Figure 19: Irregular plates controlled by α .



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Z. Huang, L. Peng, *An Improved Plate Deep Energy Method for the Bending, Buckling and Free Vibration Problems of Irregular Kirchhoff Plates*, *Engineering Structures*, Vol. 301, 117235, 2024.



Q&A

Thank you for listening!
Your feedback will be highly appreciated!